

Projective maximal families of orthogonal measures and global Σ -uniformization

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Abstract

We combine the Fischer–Friedman–Törnquist construction of projective maximal orthogonal families of measures with the global Σ -uniformization method. The point of the argument is that the Fischer–Friedman coding machinery which produces the projective wellorder can also be used as the base coding predicate in the uniformization construction. Working over L , we obtain two models. In the first one, $\mathfrak{b} = \mathfrak{c} = \omega_3$; in the second one, $\mathfrak{b} = \omega_1$ and $\mathfrak{c} = \omega_2$. In both models there is a Π_2^1 maximal orthogonal family of Borel probability measures on 2^ω , there are no boldface Σ_2^1 maximal orthogonal families, and the Σ_n^1 -uniformization property holds for every $n \geq 2$.

Keywords. Projective uniformization; maximal orthogonal families; projective wellorders; definable coding; cardinal characteristics.

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1 Introduction

Let $\mathcal{P}(2^\omega)$ denote the Polish space of Borel probability measures on 2^ω . Two measures $\mu, \nu \in \mathcal{P}(2^\omega)$ are orthogonal, written $\mu \perp \nu$, if there is a Borel set $B \subseteq 2^\omega$ such that $\mu(B) = 0$ and $\nu(2^\omega \setminus B) = 0$. A family $\mathcal{A} \subseteq \mathcal{P}(2^\omega)$ is a maximal orthogonal family, or an m.o. family, if its elements are pairwise orthogonal and it is maximal with respect to this property.

The descriptive complexity of maximal orthogonal families is closely connected to the structure of the real line. Preiss and Rataj proved that there are no analytic maximal orthogonal families [PR85]. Fischer and Törnquist proved that $V = L$ implies the existence of a coanalytic maximal orthogonal family [FT10]. Fischer–Friedman–Törnquist later showed that Π_2^1 maximal orthogonal families are compatible with large continuum and with a Δ_3^1 wellorder of the reals. More precisely, they produced models in which there is a Π_2^1 maximal orthogonal family, no boldface Σ_2^1 maximal orthogonal family, and either $\mathfrak{b} = \mathfrak{c} = \omega_3$ or $\mathfrak{b} = \omega_1, \mathfrak{c} = \omega_2$; see [FFT12]. This belongs to the broader line of work on definable maximal families, projective wellorders and cardinal characteristics developed in, among others, [JS70, Har77, FF10, FFZ11, FFK14b, FFK12, FFK14a].

The present paper explains that the same coding machinery can be combined with global Σ -uniformization. For a projective pointclass Γ , let Γ -UP denote the assertion that every set $A \subseteq 2^\omega \times 2^\omega$ in Γ admits a uniformization whose graph is again in Γ . We write

$$\text{UP}_\Sigma \quad \text{for} \quad \forall n \geq 2 \ (\Sigma_n^1\text{-UP}).$$

The case $n = 2$ follows from Kondo’s theorem; see for instance [Kec95, Mos09]. Thus the forcing construction has to arrange the cases $n \geq 3$. Classical routes to projective uniformization go either through good projective wellorders, beginning with Addison’s theorem [Add59], or through determinacy and large cardinals, as in the work of Moschovakis, Martin–Steel and Woodin [Mos71, MS89, Woo82].

The main results are the following two consistency theorems.

Theorem 1.1 (large continuum). *Assume $V = L$. There is a forcing extension in which*

$$\mathfrak{b} = \mathfrak{c} = \omega_3,$$

there is a Π_2^1 maximal orthogonal family of Borel probability measures on 2^ω , there are no boldface Σ_2^1 maximal orthogonal families, and UP_Σ holds.

Theorem 1.2 (small bounding number). *Assume $V = L$. There is a forcing extension in which*

$$\mathfrak{b} = \omega_1, \quad \mathfrak{c} = \omega_2,$$

there is a Π_2^1 maximal orthogonal family of Borel probability measures on 2^ω , there are no boldface Σ_2^1 maximal orthogonal families, and UP_Σ holds.

The relevant new point is not a new construction of maximal orthogonal families. We use the Fischer–Friedman–Törnquist construction for that part. The point is rather that the coding apparatus of Fischer–Friedman can be used as the base Σ_3^1 predicate for a global uniformization construction. In earlier global-uniformization arguments, the base predicate was obtained from a separate coding machinery. Here the base predicate is extracted from the same local coding coordinates which already underlie the projective wellorder and the projective definition of the maximal orthogonal family in [FFT12, FFK14b].

This paper therefore sits inside the author’s program on forcing separation, reduction and uniformization in the projective hierarchy without appealing to projective determinacy. The lower-level separation and reduction side is developed in [Hof22, Hof23b, Hof23a, Hof26b]; the forcing-axiom and global- Σ side is developed in [Hof24, Hof25a, Hof25d]; and the Π -uniformization and mixed Π/Σ directions are treated in [Hof21a, Hof25c, Hof26a, Hof26c]. The local-Delfino work [HM26] is complementary: it studies regularity and local uniformization phenomena in a setting where the global Σ -side and the local Π -side pull in opposite directions.

The proof has a modular form. First, we isolate a Fischer–Friedman coding package. It gives a uniformly Σ_3^1 one-variable predicate $\Phi_{\text{FF}}(z)$ saying that the namespace-tagged real z has been intentionally written into the fixed Fischer–Friedman coding apparatus. The actual coding coordinate is construction bookkeeping: at a positive coding stage we choose a fresh coordinate and force with the corresponding Fischer–Friedman coding poset, but this coordinate is not an argument of the final projective predicate. The package has intentional coding and a no-accidental-coding theorem. Moreover, it has two implementations: a finite-support σ -centered implementation, used for the model with $\mathfrak{b} = \mathfrak{c} = \omega_3$, and a countable-support S -proper implementation, used for the model with $\mathfrak{b} = \omega_1$ and $\mathfrak{c} = \omega_2$.

Second, we run the global Σ -uniformization bookkeeping using Φ_{FF} in place of the earlier coding predicate. At a uniformization stage, one lists triples (x, y, a_0) and chooses the least successful triple. Coherent name-level compression maps turn the information about the earlier failures, or about the later exclusions, into a single real. The corresponding uniformization payload is then either intentionally coded into the Fischer–Friedman apparatus, or the stage is trivial and that payload is not intentionally coded at all. The final projective definition of the uniformizing graph selects precisely the least successful y , using the no-unwanted-codes theorem to read omitted tags as negative information.

Finally, the maximal-orthogonal-family bookkeeping and the uniformization bookkeeping are interleaved on disjoint coding reservoirs and disjoint real-payload namespaces. Disjointness of both, together with the no-accidental-coding lemma, ensures that the two parts do not interfere with each other.

2 Preliminaries on maximal orthogonal families

We collect the facts about orthogonal measures which are used in the forcing argument. The aim of this section is not to reprove the full Fischer–Friedman–Törnquist construction, but to

make explicit which parts of that construction will be used as black boxes and which parts are preserved when the new uniformization stages are inserted.

Let $X = 2^\omega$ and let $\mathcal{P}(X)$ denote the Polish space of Borel probability measures on X , equipped with the usual weak topology. We write $\mu \perp \nu$ if μ and ν are mutually singular, and we write $\mu \not\perp \nu$ otherwise. Thus $\mu \perp \nu$ means that there is a Borel set $B \subseteq X$ such that

$$\mu(B) = 0 \quad \text{and} \quad \nu(X \setminus B) = 0.$$

A family $\mathcal{A} \subseteq \mathcal{P}(X)$ is orthogonal if any two distinct members of $\mathcal{A}$ are mutually singular. It is a maximal orthogonal family if it is orthogonal and every $\mu \in \mathcal{P}(X)$ is non-orthogonal to at least one member of $\mathcal{A}$. Since point measures form a closed pairwise orthogonal family, and every point measure is orthogonal to every non-atomic measure, the forcing constructions usually build the non-atomic part of the family and then add the point measures at the end. Atomic measures are then covered by the point masses at their atoms.

We shall use the following elementary descriptive facts several times.

Lemma 2.1 (basic definability of orthogonality). *The relation*

$$\{(\mu, \nu) \in \mathcal{P}(X)^2 : \mu \perp \nu\}$$

is Borel. In fact, if $(C_i : i < \omega)$ is a fixed enumeration of the clopen subsets of 2^ω , then

$$\mu \perp \nu \iff \forall k \exists i (\mu(C_i) < 2^{-k} \wedge \nu(X \setminus C_i) < 2^{-k}).$$

Consequently, for every projective pointclass Γ , the relations “ $\mu \perp \nu$ ”, “ $\mu \not\perp \nu$ ”, and “ ν is non-atomic” may be inserted into a Γ -definition without changing the projective level.

Proof. For clopen C , the map $\mu \mapsto \mu(C)$ is continuous on $\mathcal{P}(X)$, so the displayed right-hand side is Borel. If $\mu \perp \nu$, choose a Borel set B with $\mu(B) = 0$ and $\nu(X \setminus B) = 0$. By regularity, for every k one may find a compact $K \subseteq B$ with $\nu(K) > 1 - 2^{-k}$ and an open $U \supseteq B$ with $\mu(U) < 2^{-k}$. Since 2^ω is zero-dimensional, there is a clopen C with $K \subseteq C \subseteq U$. Then $\mu(C) < 2^{-k}$ and $\nu(X \setminus C) < 2^{-k}$.

Conversely, suppose such clopen sets C_k exist for all k . Replacing the bounds by a summable subsequence if necessary, put

$$B = \bigcup_m \bigcap_{k \geq m} C_k.$$

Then $\mu(B) = 0$, while

$$X \setminus B \subseteq \bigcap_m \bigcup_{k \geq m} (X \setminus C_k),$$

and the latter sets have ν -measure tending to 0. Thus $\nu(X \setminus B) = 0$, and hence $\mu \perp \nu$. $\square$

Lemma 2.2 (countability of local intersections). *Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an orthogonal family and let $\mu \in \mathcal{P}(X)$. Then*

$$\mathcal{A}_\mu = \{\nu \in \mathcal{A} : \nu \not\perp \mu\}$$

is countable.

Proof. For each $\nu \in \mathcal{A}_\mu$, let

$$\nu = f_\nu \mu + \nu^\perp$$

be the Lebesgue decomposition of ν with respect to μ . Since $\nu \not\perp \mu$, the absolutely continuous part is non-zero, so $f_\nu > 0$ on a set of positive μ -measure. If $\nu, \rho \in \mathcal{A}_\mu$ are distinct, then

$\nu \perp \rho$, and therefore their absolutely continuous parts $f_\nu \mu$ and $f_\rho \mu$ are mutually singular. Equivalently, $f_\nu f_\rho = 0$ μ -almost everywhere.

For $n, m < \omega$, let

$$A_{n,m} = \{\nu \in \mathcal{A}_\mu : \mu(\{x : f_\nu(x) > 2^{-n}\}) > 2^{-m}\}.$$

The sets $\{x : f_\nu(x) > 2^{-n}\}$ are pairwise μ -almost disjoint as ν varies through a fixed $A_{n,m}$. Hence $A_{n,m}$ is finite. Since every $\nu \in \mathcal{A}_\mu$ belongs to some $A_{n,m}$, the family $\mathcal{A}_\mu$ is countable. $\square$

The obstruction to low-complexity maximal orthogonal families is obtained from the turbulence argument of Preiss–Rataj and Fischer–Törnquist. We recall the form which is used in the two final forcing extensions.

Proposition 2.3 (Fischer–Friedman–Törnquist). *Let $a \in 2^\omega$. If there is either a Cohen real over $L[a]$ or a random real over $L[a]$, then there is no $\Sigma_2^1(a)$ maximal orthogonal family of Borel probability measures on 2^ω .*

Proof. This is [FFT12, Proposition 1]. We recall the mechanism because it explains exactly what has to be preserved by our iterations.

There is a continuous map $x \mapsto \mu^x$ from a Polish parameter space into the non-atomic probability measures on 2^ω such that the induced equivalence relation

$$x E_I y \iff \mu^x \not\perp \mu^y$$

is the turbulent equivalence relation used by Fischer–Törnquist. Equivalently, by Kakutani’s dichotomy for the relevant product measures, $x E_I y$ says that μ^x and μ^y are equivalent rather than mutually singular. The E_I -classes are meager, and the relation has the usual generic ergodicity consequences for Baire measurable, respectively Lebesgue measurable, assignments into countable structures.

Suppose toward a contradiction that $\mathcal{A}$ is a $\Sigma_2^1(a)$ maximal orthogonal family. For each x , maximality gives at least one $\nu \in \mathcal{A}$ with $\nu \not\perp \mu^x$, while Lemma 2.2 shows that the section

$$\mathcal{A}_x = \{\nu \in \mathcal{A} : \nu \not\perp \mu^x\}$$

is countable. The relation defining the pairs (x, ν) with $\nu \in \mathcal{A}_x$ is still $\Sigma_2^1(a)$ by Lemma 2.1. The argument of [FFT12], using the standard countable-section analysis for $\Sigma_2^1(a)$ relations, produces from this a projective countable-valued assignment which is invariant on E_I -classes.

If there is a Cohen real over $L[a]$, the relevant $\Sigma_2^1(a)$ and $\Delta_2^1(a)$ sets have the Baire property in the sense needed for the turbulence argument. If there is a random real over $L[a]$, they have the corresponding Lebesgue measurability property. Generic ergodicity of E_I then forces the countable-valued invariant assignment to be generically constant. This is incompatible with the construction of the product measures: a fixed countable set of measures cannot meet the non-orthogonality section of μ^x for a comeager, respectively conull, set of parameters x . This contradiction rules out $\mathcal{A}$. $\square$

We shall use the preceding proposition only through the following preservation observation.

Lemma 2.4 (persistence of the obstruction). *Let a be a real in an intermediate forcing extension.*

- (i) *If a later stage adds a Cohen real over that intermediate extension, then the final extension contains a Cohen real over $L[a]$.*
- (ii) *If a later stage adds a random real over that intermediate extension, then the final extension contains a random real over $L[a]$.*

Therefore, in either case, the final extension has no $\Sigma_2^1(a)$ maximal orthogonal family.

Proof. If c is Cohen over the intermediate extension, then c meets every dense open subset of $2^{<\omega}$ belonging to $L[a]$, since all those dense sets already belong to the intermediate model. This remains true in every further outer model. The random case is the same: being random over $L[a]$ means avoiding every Borel null set coded in $L[a]$, and the collection of such codes is fixed once a is fixed. Proposition 2.3 gives the last assertion. $\square$

We now state the precise form of the Fischer–Friedman–Törnquist construction which will be imported. The original construction has two versions, corresponding to the two cardinal-characteristic patterns that we use later. In both versions the construction is organized by local coding coordinates. A block is reserved for one task, used at most once, and then never reused. Membership in the final maximal orthogonal family is read from those blocks by a Π_2^1 definition.

Proposition 2.5 (Fischer–Friedman–Törnquist). *There are forcing constructions over L with the following properties.*

- (i) *A finite-support σ -centered construction of length ω_3 produces a model with*

$$\mathfrak{b} = \mathfrak{c} = \omega_3,$$

a Π_2^1 maximal orthogonal family, and no boldface Σ_2^1 maximal orthogonal families.

- (ii) *A countable-support S -proper construction of length ω_2 produces a model with*

$$\mathfrak{b} = \omega_1, \quad \mathfrak{c} = \omega_2,$$

a Π_2^1 maximal orthogonal family, and no boldface Σ_2^1 maximal orthogonal families.

Moreover, in both constructions the coding can be arranged on pairwise disjoint local blocks, and the following structural features hold.

- (a) **Bookkeeping for measures.** *Every name for a non-atomic Borel probability measure is considered cofinally often. When a stage sees that the interpreted measure is orthogonal to the family constructed so far, it inserts a measure equivalent to it and codes the inserted measure into a fresh block. If the measure is already non-orthogonal to an earlier member, the stage does not add a new member of the family.*
- (b) **Preservation of orthogonality.** *The measure inserted at a successful stage is chosen inside the equivalence class of the measure currently under consideration, and the coding forcing used at that stage preserves the orthogonality relations to all earlier inserted measures.*
- (c) **Projective definition.** *In the final extension, membership in the constructed family is given by a Π_2^1 formula. Informally, a measure belongs to the family iff every countable suitable transitive model containing the relevant code sees a correct local coding block witnessing that the measure was inserted at a successful stage.*
- (d) **No unwanted membership codes.** *If a block has not been intentionally used for the maximal-orthogonal-family construction, then later forcing does not make it look as if it had been used. Thus the Π_2^1 definition reads exactly the intended members of the family.*

Proof. The two forcing constructions are Theorems 1 and 2 of [FFT12]. The finite-support version uses the σ -centered almost-disjoint coding developed in the Fischer–Friedman projective-wellorder constructions. The countable-support version uses the S -proper coding machinery from [FFK14b]. We spell out why the listed structural features are the ones needed later.

For (a), the construction enumerates names for measures. At a successful stage, the current measure name is interpreted in the current intermediate model; if it is orthogonal to the previous family, an equivalent measure is selected and coded. Since equivalent measures have the same orthogonality relation to all previously constructed measures, this is the right operation for maximality.

For (b), the forcing used to add the code is chosen so as not to destroy the already verified singularity witnesses between previously inserted measures. This is part of the preservation analysis of the original Fischer–Friedman–Törnquist construction. In the finite-support case the relevant coding is σ -centered and hence c.c.c.; in the countable-support case the iterands are S -proper and satisfy the preservation properties needed for the measure-coding argument.

For (c), the point is that the code for membership is local. The assertion that a real codes a successful insertion stage can be checked inside countable suitable transitive models which contain the relevant parameters and compute the local block correctly. The external definition is therefore universal over real codes for such countable models, followed by an arithmetical verification inside the coded model; this yields a Π_2^1 definition.

For (d), the original no-unwanted-coding theorem says that a local coding pattern required by the Π_2^1 membership definition can appear only when the construction has deliberately produced the corresponding successful-insertion code. Thus a later stage cannot accidentally create a membership code on a coordinate which was not used by the maximal-orthogonal-family bookkeeping. $\square$

Remark 2.6 (why disjoint reservoirs and namespaces suffice). In the final constructions we reserve one class of Fischer–Friedman blocks for the maximal orthogonal family and a disjoint class for the global Σ -uniformization bookkeeping. Since the final predicate $\Phi_{\text{FF}}(z)$ is one-variable, we also keep the real payloads themselves in disjoint recursive namespaces. Maximal-orthogonal-family stages code only MO-records, while uniformization stages code only odd or even uniformization records. Proposition 2.5 is stable under this separation. The maximality proof only needs cofinally many stages which inspect all measure names and have access to fresh maximal-orthogonal-family blocks. The projective definition only reads successful-insertion records in the MO namespace. Therefore additional forcing stages which use disjoint blocks and disjoint payload namespaces can affect neither the membership definition of the family nor the absence of boldface Σ_2^1 maximal orthogonal families, provided that the Cohen or random stages required by Lemma 2.4 still occur after every real parameter.

Proposition 2.7 (maximal-orthogonal-family module under side forcing). *Consider either the finite-support or the countable-support Fischer–Friedman–Törnquist maximal-orthogonal-family construction, and interleave it with additional stages satisfying the following requirements. The maximal-orthogonal-family stages remain locally exhaustive for measure names and use fresh coordinates from $\mathcal{B}^{\text{MO}}$ together with MO-namespace insertion records. Every side stage is in the corresponding Fischer–Friedman preservation class, uses no coordinate from $\mathcal{B}^{\text{MO}}$, and codes no MO-namespace insertion record. Finally, the Cohen, respectively random, stages used to rule out boldface Σ_2^1 maximal orthogonal families still occur after every real parameter.*

Then the mixed extension has the same Π_2^1 maximal orthogonal family supplied by the Fischer–Friedman–Törnquist construction, and it has no boldface Σ_2^1 maximal orthogonal family.

Proof. The proof is exactly the verification isolated in Proposition 2.5 and Remark 2.6. Orthogonality and maximality use only the locally exhaustive measure bookkeeping and the preservation theorem for the relevant Fischer–Friedman implementation. The Π_2^1 definition reads only successful-insertion records in the MO-namespace on $\mathcal{B}^{\text{MO}}$ -coordinates, while side stages use disjoint coordinates and disjoint payload namespaces; the no-unwanted-codes theorem therefore prevents accidental membership records. The absence of boldface Σ_2^1 maximal orthogonal families follows from Lemma 2.4, since by hypothesis the required Cohen or random real appears after each real parameter. $\square$

3 The Fischer–Friedman coding package

The coding facts needed in the sequel are isolated in this section. The formulation follows the coding architecture used in the large-continuum global-uniformization construction: the final projective predicate is a one-variable predicate saying that a real has been written into the fixed coding apparatus. The auxiliary coding coordinate used by the forcing is not an argument of the projective predicate. It is only a resource chosen by the bookkeeping in order to make the no-unwanted-codes argument work.

Thus negative information is not produced by an additional forcing operation on a fresh block. It is obtained in the usual way: if a namespace-tagged real is never intentionally coded by the iteration, then the no-unwanted-codes theorem implies that it is not coded in the final extension. This is the convention used throughout the paper.

Coding locations and reservoirs

The underlying apparatus consists of a definable independent sequence $\vec{S} = \langle S_\alpha : \alpha < \omega_1 \rangle$ of Suslin trees, together with the Fischer–Friedman/Fischer–Friedman–Törnquist localization and almost-disjoint coding machinery. A positive coding stage is externally described as follows. Given a real z and a fresh coding coordinate η , the forcing

$$\mathbb{C}_{\text{FF}}(z, \eta)$$

uses the η -th reservoir column to select an almost disjoint set of ω -blocks of $\vec{S}$, writes the bits of z into those blocks by choosing one tree from each pair, and then almost-disjointly codes the resulting set of ordinals into a real witness. This is the same mechanism as in the large-continuum construction: if $g_\eta \subseteq \omega_1$ is the relevant ω_1 -Cohen column and $\rho : ([\omega_1]^\omega)^L \rightarrow \omega_1$ is the fixed constructible bijection, one uses

$$h_\eta = \{\rho(g_\eta \cap \alpha) : \alpha < \omega_1\}$$

as the set of starting points of the ω -blocks on which the bits of z are read. Different fresh coordinates give the required almost disjointness of the coding areas. The actual coding coordinate η , and the corresponding set h_η , are not parameters of the final projective predicate.

Convention 3.1 (coding reservoirs). *We reserve disjoint definable classes of such coding coordinates*

$$\mathcal{B}^{\text{MO}} \quad \text{and} \quad \mathcal{B}^\Sigma.$$

Coordinates from $\mathcal{B}^{\text{MO}}$ are used by the maximal-orthogonal-family bookkeeping. Coordinates from $\mathcal{B}^\Sigma$ are used by the global Σ -uniformization bookkeeping. A positive stage always uses a fresh coordinate from the appropriate reservoir. If a stage is supposed to keep a tag uncoded, the stage is simply trivial and uses no coding coordinate.

Convention 3.2 (payload namespaces). *Fix once and for all a recursive coding of finite tuples of reals and natural numbers by reals. We use it with pairwise disjoint recursive namespaces, obtained by putting a fixed marker as the first coordinate. The namespaces used in this paper are*

$$\text{MO}, \quad \text{U}_{\text{odd}}, \quad \text{U}_{\text{even}}, \quad \text{D}.$$

Thus a maximal-orthogonal-family payload has the form

$$\langle \text{MO}, \text{successful-insertion data} \rangle,$$

an odd uniformization payload has the form

$$\langle \text{U}_{\text{odd}}, e, p, x, y, a_0, b_1, \dots, b_{2q-2} \rangle,$$

and an even uniformization payload has the form

$$\langle \text{U}_{\text{even}}, e, p, x, y, a_0, b_1, \dots, b_{2q-3} \rangle.$$

The namespace D is reserved for harmless auxiliary or dummy records, should such records be needed in parallel bookkeeping; it is not read by the definitions of the maximal orthogonal family or the uniformizing relations below. The fixed first tuple $(x, 0, 0)$ used later is not itself a special coding namespace; when it is tested, it is placed inside either the U_{odd} - or the U_{even} -namespace. The ranges of these tuple maps are disjoint. Consequently, because Φ_{FF} is a one-variable predicate, coding a real in one namespace can never make a tag in another namespace count as coded.

Definition 3.3 (FF-respecting iterations). *An iteration is FF-respecting if it satisfies the following requirements.*

- (1) *Every positive coding stage uses an iterand of the form $\mathbb{C}_{\text{FF}}(\dot{z}, \dot{\eta})$, where $\dot{z}$ is a namespace-tagged real of the module currently being run, and $\dot{\eta}$ is forced to be a fresh coordinate from the appropriate reservoir.*
- (2) *Distinct positive coding stages use distinct coding coordinates. If the same mathematical tag is presented again and has already been coded, no new coding is needed. If the case distinction says that the tag must remain uncoded, the stage is trivial.*
- (3) *The set of intentionally coded reals is exactly the set of namespace-tagged payloads z which occur at positive coding stages. A maximal-orthogonal-family stage may use only the MO -namespace, and a uniformization stage may use only the U_{odd} - or U_{even} -namespace.*
- (4) *All side iterands belong to the preservation class covered by the relevant Fischer–Friedman no-unwanted-codes theorem. In the finite-support implementation these are the finite-support Fischer–Friedman–Törnquist coding iterands together with the Cohen and Hechler stages used later. In the countable-support implementation these are the corresponding S -proper, ω^ω -bounding iterands and the random stages used later.*

The last clause is deliberately restrictive. We do not use a preservation theorem for arbitrary c.c.c. or arbitrary proper forcing; we use only the concrete stages which occur in the two final constructions and which are covered by the original preservation arguments. In the finite-support construction this means the Fischer–Friedman coding pieces, Cohen forcing and Hechler forcing. Cohen forcing is countable and Hechler forcing is σ -centered, so they are among the finite-support side forcings for which the unused-tree/no-unwanted-code preservation argument applies. In the countable-support construction this means the S -proper, ω^ω -bounding

Fischer–Friedman coding pieces together with the random stages allowed by the corresponding countable-support preservation theorem. Thus the phrase “preservation class” always refers to these concrete imported preservation results, not to a general closure theorem for all c.c.c. or all proper forcing.

The projective coding predicate

Definition 3.4 (the base FF-code predicate). *For a real z , let*

$$\Phi_{\text{FF}}(z)$$

mean that z is coded into the Fischer–Friedman apparatus. More explicitly, $\Phi_{\text{FF}}(z)$ is the assertion that there is a real r such that every countable transitive suitable model M containing r, z , computing ω_1^L , the almost disjoint family and the relevant initial segment of $\vec{S}$ correctly, sees in $L[r]^M$ a coded transitive ZFC^- -model N of size $\aleph_1^M$ and an $\aleph_1^M$ -sized set of block-starting ordinals γ such that, for every $n \in \omega$,

$$n \in z \iff N \models “S_{\omega_{\gamma+2n+1}} \text{ has an } \omega_1\text{-branch}”,$$

and

$$n \notin z \iff N \models “S_{\omega_{\gamma+2n}} \text{ has an } \omega_1\text{-branch}”.$$

This is the usual local formulation of “ z is written into $\vec{S}$ ”. The witness r contains the almost-disjoint code for the relevant set of ordinals, but the external coding coordinate used to create that set is not part of the displayed predicate.

Lemma 3.5 (complexity of the FF predicate). *The relation $\Phi_{\text{FF}}(z)$ is Σ_3^1 , uniformly in the real z .*

Proof. The formula has the form $\exists r \Psi(r, z)$, where $\Psi(r, z)$ is the universal statement over real codes for countable suitable transitive models described in Definition 3.4. Correctness of the collapse, correctness of the internal construction of the almost-disjoint family, and correctness of the initial segment of $\vec{S}$ are arithmetical or coanalytic in the code for the countable model, and the remaining verification is first-order over the decoded structures. This is exactly the complexity calculation in the large-continuum coding construction and in the Fischer–Friedman coding machinery. Hence Φ_{FF} is Σ_3^1 . $\square$

Intentional coding and no unwanted codes

Lemma 3.6 (positive coding). *Let η be a fresh coding coordinate and let $\dot{z}$ be a name for a real. The forcing $\mathbb{C}_{\text{FF}}(\dot{z}, \eta)$ can be chosen so that*

$$\mathbb{C}_{\text{FF}}(\dot{z}, \eta) \Vdash \Phi_{\text{FF}}(\dot{z}).$$

In the finite-support implementation this is the σ -centered final coding piece of the Fischer–Friedman apparatus. In the countable-support implementation it is the corresponding S -proper coding piece.

Proof. The forcing chooses the almost disjoint family of ω -blocks determined by the fresh coordinate η , writes the bits of $\dot{z}$ into the corresponding pairs of Suslin trees, and almost-disjointly codes the resulting set of ordinals into a real witness r . The local verification of Definition 3.4 is then exactly the usual David-trick/local-model verification from the large-continuum construction. The preservation theorem for the relevant implementation keeps this witness valid in all later FF-respecting extensions. $\square$

Lemma 3.7 (omission and no unwanted codes). *Let $\mathbb{P}$ be an FF-respecting iteration and let $G \subseteq \mathbb{P}$ be generic. Let*

$$X_G = \{\dot{z}_\alpha^G : \alpha \text{ is a positive coding stage of } \mathbb{P}\}$$

be the set of intentionally coded namespace-tagged reals. If $z \notin X_G$, then

$$L[G] \models \neg \Phi_{\text{FF}}(z).$$

Thus a construction stage which is supposed to make a tag negative does not close or reserve a block; it simply never intentionally codes that namespace-tagged real.

Proof. This is the no-unwanted-codes argument in its negative form. We recall the finite-support proof because it is the one used in the large-continuum construction. Suppose toward a contradiction that $\Phi_{\text{FF}}(z)$ holds, witnessed by a real r , although $z \notin X_G$. The reals z, r belong to a countable subiteration together with only countably many of the preliminary ω_1 -Cohen columns and the coding coordinates used by that subiteration. Since the intentionally coded reals in this submodel use almost disjoint families of ω -blocks, one can choose a block-starting ordinal γ and a bit n at which the local pattern required by the alleged witness r demands a branch through a Suslin tree which none of the forcings in the submodel has added. Independence of $\vec{S}$, preservation of unused Suslin trees by the almost-disjoint coding forcings, and preservation by the allowed side forcings imply that this tree is still Suslin in the submodel. Here the allowed side forcings are exactly the ones named in Definition 3.3: in the finite-support case, Cohen and Hechler forcing are covered by the same finite-support preservation proof, and no arbitrary c.c.c. forcing is being used. Hence no real in the submodel can have an $L[\cdot]$ which sees such a branch. This contradicts the local consequence of $\Psi(r, z)$.

The countable-support version is the corresponding no-unwanted-codes theorem for the S -proper Fischer–Friedman implementation. In that case the random stages are among the side stages permitted by the countable-support preservation theorem, and the present iteration uses no other side forcing. Hence the same conclusion applies. $\square$

Lemma 3.8 (no accidental FF codes). *Let $\mathbb{P}$ be an FF-respecting iteration and $G \subseteq \mathbb{P}$ generic. Then in $L[G]$,*

$$\{z : \Phi_{\text{FF}}(z)\} = X_G,$$

where X_G is the set of intentionally coded namespace-tagged reals. In particular, stages using $\mathcal{B}^{\text{MO}}$ and stages using $\mathcal{B}^\Sigma$ do not create codes for one another, because their intentionally coded payloads are drawn from disjoint recursive namespaces.

Proof. The inclusion $X_G \subseteq \{z : \Phi_{\text{FF}}(z)\}$ is Lemma 3.6. The reverse inclusion is Lemma 3.7. The final assertion is just the disjointness of the namespace markers included in the real coded at positive stages; the projective predicate sees the namespace-tagged real itself, not an external coding coordinate. $\square$

Lemma 3.9 (FF coding package). *The predicate Φ_{FF} and the forcings $\mathbb{C}_{\text{FF}}(z, \eta)$ may be used in either of the final constructions with the following properties.*

- (1) Φ_{FF} is a uniformly Σ_3^1 predicate of one real.
- (2) For every namespace-tagged real z and every fresh coordinate η in the appropriate reservoir, $\mathbb{C}_{\text{FF}}(z, \eta)$ forces $\Phi_{\text{FF}}(z)$.
- (3) If a namespace-tagged real is not intentionally coded at any stage, then it satisfies $\neg \Phi_{\text{FF}}$ in the final extension.

- (4) *FF-respecting iterations have no accidental codes, and the block reservoirs together with the payload namespaces do not interfere.*
- (5) *In the large-continuum construction the coding forcings fit into a finite-support c.c.c. iteration; the final coding pieces are σ -centered.*
- (6) *In the $\mathfrak{b} = \omega_1$, $\mathfrak{c} = \omega_2$ construction the coding forcings fit into the countable-support S -proper, ω^ω -bounding iteration used by the Fischer–Friedman–Törnquist construction.*

Proof. Items (1)–(4) are Lemmas 3.5, 3.6, 3.7 and 3.8. Items (5) and (6) are the two implementations of the Fischer–Friedman coding apparatus used in the final sections. The important restriction is that the iteration uses only the side forcings covered by the corresponding preservation theorem. $\square$

Remark 3.10 (why the abstraction is sufficient). The later proof uses exactly the package above. Positive coding supplies the base Σ_3^1 predicate for odd uniformization levels. The negative predicate used at even levels is simply $\neg\Phi_{\text{FF}}$, and it is forced by omission together with the no-unwanted-codes theorem. The construction uses only positive coding operations and trivial stages; no additional negative-coding operation is part of the argument.

4 Name compression and the uniformization module

We now isolate the part of the argument which turns a robust Σ_3^1 coding predicate into uniformization at all projective Σ -levels. The point of this section is to make explicit what is used later from the global Σ -uniformization construction: local candidate orders, coherent compression of predecessor data, and the four parity-dependent copying cases. The predicate Φ_{FF} from Section 3 plays the role of the base level-three coding predicate.

Throughout this section we write

$$\text{Coded}(t)$$

for the one-variable assertion $\Phi_{\text{FF}}(t)$. Here t is always a uniformization payload in one of the two namespaces from Convention 3.2: odd certificates use the marker $\mathbf{U}_{\text{odd}}$, and even certificates use the marker $\mathbf{U}_{\text{even}}$. Thus the real t contains the parity namespace, the formula number, the real parameters, the local candidate and the auxiliary compression reals. It does not contain an external coding coordinate. When the forcing construction says that it codes such a tag, it means that it chooses a fresh coordinate η from the Σ -reservoir and forces with $\mathbb{C}_{\text{FF}}(t, \eta)$. When the construction says that the tag remains uncoded, the stage is trivial. Permanence of the negative assertion is supplied by Lemma 3.8: a uniformization tag which is never intentionally coded does not satisfy Coded in the final model. Since the payload namespace is part of t , an MO-record coded for the maximal orthogonal family is never confused with a uniformization tag.

Normal forms and local candidate orders

Fix recursive enumerations of projective formulas in normal form. For odd levels we write a Σ_{2q+1}^1 formula, $q \geq 1$, as

$$\Psi_e^{2q+1}(p, x, y) \iff \exists a_0 \forall a_1 \exists a_2 \cdots \exists a_{2q-2} \psi_e^\Pi(p, x, y, a_0, a_1, \dots, a_{2q-2}),$$

where $\psi_e^\Pi \in \Pi_2^1$. When $q = 1$, the string $a_1, \dots, a_{2q-2}$ is empty and the displayed tail is simply $\psi_e^\Pi(p, x, y, a_0)$.

For even levels ≥ 4 , write a Σ_{2q}^1 formula, $q \geq 2$, as

$$\Psi_e^{2q}(p, x, y) \iff \exists a_0 \forall a_1 \exists a_2 \cdots \forall a_{2q-3} \psi_e^\Sigma(p, x, y, a_0, a_1, \dots, a_{2q-3}),$$

where $\psi_e^\Sigma \in \Sigma_2^1$.

Let $\kappa \in \{\omega_2, \omega_3\}$, and let $\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha < \kappa \rangle$ be one of the iterations used in Sections 5 and 6. If $G \subseteq \mathbb{P}_\kappa$ is generic, write $G_\gamma = G \cap \mathbb{P}_\gamma$. The construction uses only local candidate lists. At an intermediate stage $\gamma < \kappa$, for fixed $(n, e, p, x) \in L[G_\gamma]$, form the candidate list as follows. The first candidate is always the fixed dummy pair $(0, 0)$, where 0 denotes the canonical zero real. Equivalently, the first triple in the list for the input x is always

$$(x, y_0^\gamma, a_0^{\gamma,0}) = (x, 0, 0).$$

After this first entry, list all remaining candidate pairs $(y, a_0) \in L[G_\gamma]$ in the canonical least-name order, omitting repetitions of $(0, 0)$, and write

$$\langle (y_\xi^\gamma, a_0^{\gamma,\xi}) : \xi < \lambda_\gamma \rangle$$

for the resulting normalized list, where $\lambda_\gamma = (2^\omega)^{L[G_\gamma]}$. This is the convention used in the global Σ -uniformization construction: if $(0, 0)$ already witnesses the section at x , then the uniformizing value is intended to be 0; otherwise the failure of the fixed first candidate is available before every genuine later candidate.

The least-name orders are chosen coherently subject to this normalization: if $\gamma < \delta$ and $p, x \in L[G_\gamma]$, then the fixed first candidate remains first, and every $\mathbb{P}_\gamma$ -name for a non-zero candidate precedes every $\mathbb{P}_\delta$ -name which is not equivalent to a $\mathbb{P}_\gamma$ -name. Hence the candidate list of $L[G_\gamma]$ is an initial segment of the candidate list of $L[G_\delta]$. We denote by

$$\prec_{n,e,p,x}$$

the direct-limit order on the final candidates. If a candidate has local index ξ at stage γ , then its $\prec_{n,e,p,x}$ -predecessors are precisely the local candidates with indices $< \xi$ in $L[G_\gamma]$. Later forcing may append new candidates, but it never inserts a new candidate below an old one.

Coherent compression maps

We follow here the compression convention from the large-continuum global Σ -uniformization construction. The maps π are not part of the final projective definition. They are auxiliary bookkeeping maps chosen in the intermediate universes of the iteration, and their only role is to compress a sequence of witnesses for earlier candidates into one real which can then be placed into a tagged coding statement.

Let

$$R_\gamma = (2^\omega)^{L[G_\gamma]} \quad \text{and} \quad \lambda_\gamma = |R_\gamma|.$$

At stage γ we fix, in $L[G_\gamma]$, a family

$$\langle \pi_\alpha^\gamma : 0 < \alpha < \lambda_\gamma \rangle$$

of bijections

$$\pi_\alpha^\gamma : (R_\gamma)^\alpha \cap L[G_\gamma] \longrightarrow R_\gamma.$$

The superscript records the intermediate universe and the subscript records the positive length of the sequence being compressed. Thus, if a candidate has local index $\alpha > 0$, a single real b may be decoded by

$$(\pi_\alpha^\gamma)^{-1}(b) = \langle b^\eta : \eta < \alpha \rangle$$

as an α -sequence indexed by the earlier local candidates. For $\alpha = 0$ no compression map is used: there are no earlier candidates, and all predecessor clauses are read as vacuous.

The families are chosen coherently as the iteration grows. More precisely, if $\gamma < \delta$, $0 < \alpha < \lambda_\gamma$, and $\vec{r} \in (R_\gamma)^\alpha \cap L[G_\gamma]$, then

$$\pi_\alpha^\delta(\vec{r}) = \pi_\alpha^\gamma(\vec{r}).$$

This is the only coherence requirement used below: an old compressed real continues to decode to the same old sequence in later intermediate universes.

Lemma 4.1 (coherent compression maps). *For the two iterations used in Sections 5 and 6, the coherent families*

$$\langle \pi_\alpha^\gamma : 0 < \alpha < \lambda_\gamma \rangle \quad (\gamma < \kappa)$$

can be fixed, as prepared bookkeeping maps, throughout the construction.

Proof. We first record the cardinal arithmetic. In the countable-support S -proper construction of length ω_2 , every proper initial segment has size at most ω_1 and preserves CH. Hence $\lambda_\gamma = \omega_1$ for every $\gamma < \omega_2$, and for each $0 < \alpha < \lambda_\gamma$ the ordinal α is countable. Therefore

$$|(R_\gamma)^\alpha \cap L[G_\gamma]| = \omega_1 = \lambda_\gamma.$$

In the finite-support c.c.c. construction of length ω_3 , the standard GCH size calculation gives $\lambda_\gamma \leq \omega_2$ for each $\gamma < \omega_3$, and after the harmless preliminary real-adding stages $\lambda_\gamma \in \{\omega_1, \omega_2\}$. Thus GCH gives

$$\lambda_\gamma^{<\lambda_\gamma} = \lambda_\gamma,$$

and hence $|(R_\gamma)^\alpha \cap L[G_\gamma]| = \lambda_\gamma$ for every $0 < \alpha < \lambda_\gamma$.

We are not using an automatic theorem that arbitrary previously chosen onto bijections extend to later enlarged domains. The maps are chosen together with the bookkeeping, as in the reference preprint. Whenever a compression of a new sequence is first required, the current coherent partial map is extended by assigning that sequence a fresh real; if necessary, the dummy real-adding stages occur before the task is presented, so that such fresh values are available. Since only λ_γ -many compression requests occur inside $L[G_\gamma]$, the cardinal arithmetic above allows the coherent partial maps to be completed, using the canonical wellorder of $L[G_\gamma]$, to bijections $\pi_\alpha^\gamma : (R_\gamma)^\alpha \cap L[G_\gamma] \rightarrow R_\gamma$. Earlier assigned values are kept fixed, which is precisely the coherence stated above. $\square$

This is the only use of cardinal arithmetic in the copying construction. Once the families π_α^γ have been fixed, the four cases below are read exactly as in the global Σ -uniformization preprint: the bookkeeping hands the construction a local candidate index α and finitely many reals b_i . If $\alpha > 0$, the stage decodes $(\pi_\alpha^\gamma)^{-1}(b_i)$ as the sequence of witnesses attached to the candidates with indices below α . If $\alpha = 0$, no π_0 is invoked and the predecessor tests are vacuous.

Lemma 4.2 (localizing small witness sequences). *Let $G \subseteq \mathbb{P}_\kappa$ be generic for one of the final iterations. Every sequence of reals of length $< \kappa$ which appears in $L[G]$ appears in some intermediate model $L[G_\delta]$, $\delta < \kappa$.*

Consequently, suppose that c is a candidate for a task (n, e, p, x) , that $c \in L[G_\gamma]$, and that c has local index $\alpha < \lambda_\gamma < \kappa$. If $\alpha = 0$, there are no predecessors and no compression is needed. If $\alpha > 0$ and, in the final extension, one has chosen α -many real witnesses attached to the predecessors of c , then after increasing γ to some $\delta < \kappa$ the entire witness sequence belongs to $L[G_\delta]$. The candidate c is still indexed by the same old initial segment of candidates, and the compressed real $\pi_\alpha^\delta(\langle b^\eta : \eta < \alpha \rangle)$ decodes in every later model to that same sequence.

Proof. The support calculation is the standard one for the two iterations. In the finite-support c.c.c. construction, a name for a real uses countably many coordinates; the union of $< \omega_3$ -many such supports has size $< \omega_3$ and is therefore bounded in ω_3 . In the countable-support S -proper construction, the same argument uses ω_2 : the union of $< \omega_2$ -many countable supports has size $< \omega_2$ and is bounded in ω_2 . Thus the whole witness sequence is already interpreted by some proper initial segment.

For $\alpha > 0$, the statement about decoding is exactly the prepared coherence of the maps π_α^γ . Once the sequence has been compressed at stage δ , the value assigned to it is kept fixed in every later prepared map. Thus the compressed real continues to decode to the same sequence. $\square$

The four copying cases

A uniformization stage is not a separate forcing which searches for a least witness. Instead, the bookkeeping presents a local task and a local candidate index. The stage then inspects only the lower-complexity matrix for the earlier local candidates. Depending on the parity of the projective level, the stage either writes one Fischer–Friedman code into a fresh Σ -reservoir block or does nothing.

First consider an odd level $2q + 1$. At a stage γ assigned to

$$(2q + 1, e, p, x, \xi, b_1, \dots, b_{2q-2}),$$

let

$$\langle (y_\eta^\gamma, a_0^{\gamma, \eta}) : \eta < \lambda_\gamma \rangle$$

be the local candidate list in $L[G_\gamma]$. If $\xi > 0$, unpack each b_i as

$$(\pi_\xi^\gamma)^{-1}(b_i) = \langle b_i^\eta : \eta < \xi \rangle.$$

If $\xi = 0$, there are no earlier candidates, no unpacking is performed, and all conditions with quantifier $\eta < \xi$ are vacuous. There are two cases.

Odd–1. If for every $\eta < \xi$,

$$\neg \psi_e^\Pi(p, x, y_\eta^\gamma, a_0^{\gamma, \eta}, b_1^\eta, \dots, b_{2q-2}^\eta)$$

holds, then the stage codes the $\mathbf{U}_{\text{odd}}$ -payload

$$t = \langle \mathbf{U}_{\text{odd}}, e, p, x, y_\xi^\gamma, a_0^{\gamma, \xi}, b_1, \dots, b_{2q-2} \rangle$$

using one fresh coding coordinate from $\mathcal{B}^\Sigma$. Thus the final model satisfies $\text{Coded}(t)$.

Odd–2. If there is $\eta < \xi$ such that

$$\psi_e^\Pi(p, x, y_\eta^\gamma, a_0^{\gamma, \eta}, b_1^\eta, \dots, b_{2q-2}^\eta)$$

holds, the stage is trivial. Freshness of the namespace-tagged payloads and the no-accidental-coding lemma ensure that this presentation does not contribute a code for t .

For an even level $2q$, $q \geq 2$, the roles of coding and non-coding are dual. At a stage γ assigned to

$$(2q, e, p, x, \xi, b_1, \dots, b_{2q-3}),$$

use the same local candidate list. If $\xi > 0$, unpack the b_i 's by π_ξ^γ ; if $\xi = 0$, no unpacking is performed and the predecessor tests are vacuous.

Even-1. If for every $\eta < \xi$,

$$\neg \psi_e^\Sigma(p, x, y_\eta^\gamma, a_0^{\gamma, \eta}, b_1^\eta, \dots, b_{2q-3}^\eta)$$

holds, the stage is trivial. The corresponding U_{even} -payload is thereby kept uncoded.

Even-2. If there is $\eta < \xi$ such that

$$\psi_e^\Sigma(p, x, y_\eta^\gamma, a_0^{\gamma, \eta}, b_1^\eta, \dots, b_{2q-3}^\eta)$$

holds, then the stage codes the U_{even} -payload

$$t = \langle U_{\text{even}}, e, p, x, y_\xi^\gamma, a_0^{\gamma, \xi}, b_1, \dots, b_{2q-3} \rangle$$

using one fresh coding coordinate from $\mathcal{B}^\Sigma$.

Only Π_2^1 - and Σ_2^1 -matrices are tested in these cases. By Shoenfield absoluteness, once the parameters have appeared in an intermediate model, the truth values used by the case distinction are not changed by later forcing.

The induced projective certificates

The copying cases are designed so that the following projective certificates hold exactly for the least successful candidate.

For odd levels, define

$$\begin{aligned} \text{Cert}_e^{2q+1}(p, x, y, a_0) &\iff \exists b_1 \forall b_2 \exists b_3 \cdots \forall b_{2q-2} \\ &\quad \text{Coded}(\langle U_{\text{odd}}, e, p, x, y, a_0, b_1, \dots, b_{2q-2} \rangle). \end{aligned}$$

For $q = 1$, this means simply

$$\text{Coded}(\langle U_{\text{odd}}, e, p, x, y, a_0 \rangle).$$

For even levels $2q \geq 4$, define

$$\begin{aligned} \text{Cert}_e^{2q}(p, x, y, a_0) &\iff \exists b_1 \forall b_2 \exists b_3 \cdots \exists b_{2q-3} \\ &\quad \neg \text{Coded}(\langle U_{\text{even}}, e, p, x, y, a_0, b_1, \dots, b_{2q-3} \rangle). \end{aligned}$$

The quantifier strings alternate in the displayed order; in the even case the last quantifier is existential because $2q - 3$ is odd.

Finally set

$$\begin{aligned} U_e^{2q+1}(p, x, y) &\iff \exists a_0 [\forall a_1 \exists a_2 \cdots \exists a_{2q-2} \psi_e^\Pi(p, x, y, a_0, a_1, \dots, a_{2q-2}) \\ &\quad \wedge \text{Cert}_e^{2q+1}(p, x, y, a_0)] \end{aligned}$$

for odd levels, and

$$\begin{aligned} U_e^{2q}(p, x, y) &\iff \exists a_0 [\forall a_1 \exists a_2 \cdots \forall a_{2q-3} \psi_e^\Sigma(p, x, y, a_0, a_1, \dots, a_{2q-3}) \\ &\quad \wedge \text{Cert}_e^{2q}(p, x, y, a_0)] \end{aligned}$$

for even levels.

Lemma 4.3 (complexity calculation). *For every $n \geq 3$, the relation U_e^n is Σ_n^1 , uniformly in e and in the real parameter p .*

Proof. The predicate Coded is Σ_3^1 , and $\neg$ Coded is Π_3^1 . The certificate prefixes above supply exactly the missing alternating quantifiers needed to reach the indicated level. The first conjunct in the definition of U_e^n is the tail of the original Σ_n^1 normal form after the first existential witness a_0 . Placing the two conjuncts under the common existential quantifier over a_0 and prenexing gives a Σ_n^1 definition. $\square$

Lemma 4.4 (copying lemma). *Let $G \subseteq \mathbb{P}_\kappa$ be generic for one of the final iterations, and work in $L[G]$. Fix a task (n, e, p, x) , and order its final candidates by $\prec_{n,e,p,x}$.*

- (1) *Suppose $n = 2q + 1$. If $c_* = (y_*, a_0^*)$ is the least candidate such that*

$$L[G] \models \forall a_1 \exists a_2 \cdots \exists a_{2q-2} \psi_e^\Pi(p, x, y_*, a_0^*, a_1, \dots, a_{2q-2}),$$

then

$$L[G] \models \text{Cert}_e^{2q+1}(p, x, y_*, a_0^*).$$

If $c = (y, a_0)$ is a candidate with $c_ \prec_{n,e,p,x} c$, then*

$$L[G] \models \neg \text{Cert}_e^{2q+1}(p, x, y, a_0).$$

- (2) *Suppose $n = 2q \geq 4$. If $c_* = (y_*, a_0^*)$ is the least candidate such that*

$$L[G] \models \forall a_1 \exists a_2 \cdots \forall a_{2q-3} \psi_e^\Sigma(p, x, y_*, a_0^*, a_1, \dots, a_{2q-3}),$$

then

$$L[G] \models \text{Cert}_e^{2q}(p, x, y_*, a_0^*).$$

If $c = (y, a_0)$ is a candidate with $c_ \prec_{n,e,p,x} c$, then*

$$L[G] \models \neg \text{Cert}_e^{2q}(p, x, y, a_0).$$

Proof. We give the odd case in detail. The even case is the same argument with coding and non-coding interchanged.

Assume $n = 2q + 1$, and let $c_* = (y_*, a_0^*)$ be the least successful candidate. Choose $\gamma_0 < \kappa$ such that $p, x, y_*, a_0^* \in L[G_{\gamma_0}]$, and let ξ be the local index of c_* in the candidate list of $L[G_{\gamma_0}]$. By coherence of the candidate orders, the predecessors of c_* are exactly

$$c_\eta = (y_\eta^{\gamma_0}, a_0^{\gamma_0, \eta}), \quad \eta < \xi,$$

and this remains true in later intermediate models.

For every $\eta < \xi$, the candidate c_η is not successful. Thus in the final extension one can choose first-level failure witnesses b_1^η for all predecessors. The sequence $\langle b_1^\eta : \eta < \xi \rangle$ has length $< \kappa$, so by Lemma 4.2 it belongs to some later $L[G_{\gamma_1}]$. Define

$$b_1 = \pi_\xi^{\gamma_1}(\langle b_1^\eta : \eta < \xi \rangle).$$

Now let b_2 be arbitrary in the final extension. Increase to a stage $\gamma_2 \geq \gamma_1$ with $b_2 \in L[G_{\gamma_2}]$, and unpack

$$(\pi_\xi^{\gamma_2})^{-1}(b_2) = \langle b_2^\eta : \eta < \xi \rangle.$$

For each predecessor c_η , use the failure of c_η to choose the next coordinatewise response b_3^η . Localize the resulting sequence, compress it to a single real b_3 , and continue through the finite alternating string. At each existential step the coordinatewise witnesses are chosen first in the final extension, then localized to a later intermediate model, and only then compressed

by the relevant π_ξ^γ . Coherence of the compression maps ensures that previously compressed reals keep the same unpacking after the stage is increased.

At the end we have produced, against the arbitrary universal choices, a tuple $(b_1, \dots, b_{2q-2})$ and a stage $\delta < \kappa$ containing all entries such that, for every $\eta < \xi$,

$$\neg \psi_e^\Pi(p, x, y_\eta^\delta, a_0^{\delta, \eta}, b_1^\eta, \dots, b_{2q-2}^\eta)$$

holds. By cofinality of the bookkeeping, a later Σ -uniformization stage presents the data

$$(2q+1, e, p, x, \xi, b_1, \dots, b_{2q-2})$$

with c_* as the local ξ -th candidate. The displayed failures put this stage in case Odd-1, and the corresponding U_{odd} -payload is deliberately Fischer–Friedman coded. Hence $\text{Cert}_e^{2q+1}(p, x, y_*, a_0^*)$ holds. For $q = 1$, this is the vacuous version of the same argument: there are no auxiliary b_i 's to compress.

Now let $c = (y, a_0)$ be a candidate with $c_* \prec_{n, e, p, x} c$. We show that the odd certificate for c fails. Fix an arbitrary first real b_1 . Choose an intermediate stage containing p, x, c_*, c, b_1 , let ξ be the local index of c , and let $\eta_* < \xi$ be the local index of c_* . The coherence of the least-name orders is used here in one essential way: once a presentation contains the later candidate c , it also contains every final predecessor of c , in particular c_* , with the same relative order. Thus the payload for c is never positively presented before the construction can see the earlier successful candidate c_* . Unpack

$$(\pi_\xi^\gamma)^{-1}(b_1) = \langle b_1^\eta : \eta < \xi \rangle.$$

Since c_* is successful, choose a response in the coordinate η_* so that the remaining alternating play can be continued to make

$$\psi_e^\Pi(p, x, y_*, a_0^*, b_1^{\eta_*}, b_2^{\eta_*}, \dots, b_{2q-2}^{\eta_*})$$

true. Fill the other coordinates arbitrarily, localize the whole sequence, compress it to the next real, and continue through the finite string. For the compressed tuple so obtained, every sufficiently late presentation of c sees the earlier local candidate c_* satisfying the Π_2^1 -matrix. Hence the stage is in case Odd-2 and is trivial. By Shoenfield absoluteness the same remains true at later presentations of the same tag. Since the corresponding uniformization payload is never deliberately coded, Lemma 3.8 gives the required failure of Coded. This supplies exactly the alternating witnesses for

$$\neg \text{Cert}_e^{2q+1}(p, x, y, a_0).$$

For even $n = 2q \geq 4$, use the same localization-before-compression argument. If c_* is least successful, the failures of all earlier candidates put sufficiently late presentations of c_* in Even-1, so the relevant uniformization payloads remain uncoded and the even certificate, which asks for $\neg$ Coded, holds. If $c_* \prec c$, the success of c_* puts the corresponding presentations of c in Even-2, so the uniformization payloads are deliberately coded and the even certificate fails. $\square$

Proposition 4.5 (uniformization module). *Let $\mathbb{P}_\kappa$ be one of the two FF-respecting iterations constructed below. Assume that the Σ -uniformization bookkeeping is cofinal for all tuples appearing in the four copying cases and that the final iteration has no accidental Fischer–Friedman codes for uniformization-namespace payloads. Then $\mathbb{P}_\kappa$ forces Σ_n^1 -uniformization for every $n \geq 3$. Consequently, together with Kondo's theorem at level 2, it forces UP_Σ .*

Proof. Work in the final extension. Let

$$A = \{(x, y) : \Psi_e^n(p, x, y)\}$$

be a Σ_n^1 relation, $n \geq 3$, with real parameter p . By Lemma 4.3, $U_e^n(p, x, y)$ is Σ_n^1 . Moreover $U_e^n(p, x, y)$ implies $\Psi_e^n(p, x, y)$, because its definition contains the corresponding tail of the normal form after some witness a_0 . Thus $U_e^n \subseteq A$.

Fix x with non-empty section A_x . Order all final candidates (y, a_0) for the task (n, e, p, x) by $\prec_{n,e,p,x}$, using the normalized list whose first triple is $(x, 0, 0)$. If this first candidate is successful, the construction selects 0 as the value on the section. If it is not successful, its failure is one of the predecessor failures used in the copying argument for every later candidate. In either case, let $c_* = (y_*, a_0^*)$ be the least successful candidate. The appropriate part of Lemma 4.4 gives the certificate for (y_*, a_0^*) . Therefore $U_e^n(p, x, y_*)$ holds, so the uniformizing relation meets the section.

For uniqueness, suppose $U_e^n(p, x, y)$ holds, witnessed by a_0 . Then (y, a_0) is a successful candidate. If it were not least among successful candidates, there would be a strictly earlier successful candidate, and the second half of Lemma 4.4 would say that the certificate for (y, a_0) fails. This contradicts the definition of U_e^n . Hence every witness to $U_e^n(p, x, y)$ is a least successful candidate, and since $\prec_{n,e,p,x}$ is linear, all such witnesses have the same y . Thus U_e^n is a Σ_n^1 -uniformization of A . $\square$

Remark 4.6. The least-name order is used only by the construction to organize local candidates. The final projective uniformizing relation does not say that y is least in that order. It says that a certain Fischer–Friedman tag is coded or not coded. The leastness is transferred into this projective certificate by the copying lemma. This is why the argument does not require a final projective wellorder of the reals.

5 The model with $\mathfrak{b} = \mathfrak{c} = \omega_3$

We now prove Theorem 1.1. This is the finite-support version of the construction. The section is written somewhat explicitly because it is the point where three independent pieces of bookkeeping are merged: the Fischer–Friedman–Törnquist maximal-orthogonal-family construction, the global Σ -uniformization module, and the Hechler/Cohen bookkeeping which controls the cardinal characteristics and rules out low-complexity maximal orthogonal families.

The mixed finite-support iteration

Work in L . First perform the preliminary Fischer–Friedman–Törnquist preparation. This preparation adds no reals. It supplies the local coding apparatus, the definable sequence of blocks from Convention 3.1, and the two disjoint reservoirs

$$\mathcal{B}^{\text{MO}} \quad \text{and} \quad \mathcal{B}^{\Sigma}.$$

The reservoir $\mathcal{B}^{\text{MO}}$ is used only by the maximal-orthogonal-family stages, and the definition of the family reads only successful-insertion records in the MO namespace. The reservoir $\mathcal{B}^{\Sigma}$ is used only by the uniformization stages, and the uniformization formulas read only $\mathcal{U}_{\text{odd}}$ - and $\mathcal{U}_{\text{even}}$ -records. Thus the payload namespaces from Convention 3.2 are used in parallel with the reservoir separation. A block assigned at one stage is never assigned again.

Let

$$I_{\text{MO}} \cup I_{\Sigma} \cup I_{\text{dom}} \cup I_{\text{Cohen}} = \omega_3$$

be a partition into four cofinal sets. We choose it locally cofinal: whenever $\gamma < \omega_3$ has uncountable cardinality, each of the four pieces has cardinality $|\gamma|$ below γ . This harmless strengthening is useful for the local compression bookkeeping, because it ensures that every initial segment of uncountable size already contains enough dummy real-adding stages and enough unused coding coordinates.

We build a finite-support iteration

$$\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha < \omega_3 \rangle.$$

The bookkeeping is the canonical L -bookkeeping. At stage α it looks at all objects which have appeared in the current intermediate model $L[G_\alpha]$ and at the least not-yet-satisfied task of the type assigned to α . Equivalently, one may fix in L bookkeeping functions which, after interpretation by G_α , enumerate $\mathbb{P}_\alpha$ -names for measure codes, projective uniformization tasks, names for functions in ω^ω , and names for real parameters. The only property needed below is the following local exhaustivity property.

For every $\beta < \omega_3$ and every object of the relevant kind in $L[G_\beta]$, there are cofinally many later stages of the appropriate type which consider that object, using a fresh block when a block is required.

The iterand is defined as follows.

- (1) If $\alpha \in I_{\text{MO}}$, the stage runs the Fischer–Friedman–Törnquist maximal-orthogonal-family procedure on the next measure name selected by the bookkeeping. In $L[G_\alpha]$, if the name interprets as a non-atomic measure which is orthogonal to the family constructed so far, the stage chooses the corresponding Fischer–Friedman–Törnquist iterand and codes the inserted equivalent measure into a fresh block from $\mathcal{B}^{\text{MO}}$. If the measure is atomic, or is already non-orthogonal to an earlier member of the family, the stage is trivial. This is exactly the finite-support σ -centered implementation isolated in Proposition 2.5.
- (2) If $\alpha \in I_\Sigma$, the stage runs one step of the uniformization bookkeeping from Proposition 4.5. The task consists of a projective formula, a real parameter, a local input x , a local candidate (y, a_0) , and one of the auxiliary compression reals required in the four copying cases of Section 4. According to the relevant case, the stage either codes the prescribed U_{odd} - or U_{even} -payload using a fresh coordinate from $\mathcal{B}^\Sigma$, or is trivial and does not intentionally code the corresponding payload. In the positive case the iterand is the finite-support σ -centered implementation of Lemma 3.9; in the negative case the iterand is the trivial forcing, and the later negative assertion is supplied by Lemma 3.8.
- (3) If $\alpha \in I_{\text{dom}}$, the iterand is Hechler forcing $\mathbb{D}$, as computed in $L[G_\alpha]$.
- (4) If $\alpha \in I_{\text{Cohen}}$, the iterand is Cohen forcing $\mathbb{C} = 2^{<\omega}$, as computed in $L[G_\alpha]$.

The phrase “as computed in $L[G_\alpha]$ ” is meant only to emphasize that the bookkeeping test is internal to the current intermediate model. The whole definition is a standard finite-support iteration by $\mathbb{P}_\alpha$ -names for posets.

Lemma 5.1. *The iteration $\mathbb{P}_{\omega_3}$ is c.c.c., preserves cardinals, has size ω_3 , and forces $\mathfrak{c} = \omega_3$.*

Proof. Every non-trivial iterand in the construction is σ -centered. This is part of the finite-support Fischer–Friedman–Törnquist implementation at the I_{MO} stages and part of the finite-support implementation of Lemma 3.9 at the I_Σ stages. Cohen forcing is countable, and Hechler forcing is σ -centered: conditions with the same finite stem form a centered set. Hence the finite-support iteration is c.c.c. Therefore cardinals and cofinalities are preserved.

The size calculation is the usual one over L . By induction on $\alpha \leq \omega_3$, each proper initial segment has size at most $\max(|\alpha|, \omega_1)$, and the full iteration has size ω_3 . The coding iterands use only fresh coordinates from a fixed L -definable apparatus, and the Cohen and Hechler iterands have size at most the current continuum; therefore no stage introduces posets of size exceeding the size of the current initial segment plus ω_1 . Since supports are finite and L satisfies GCH, the induction gives $|\mathbb{P}_{\omega_3}| = \omega_3$.

It follows that the final extension has at most ω_3 reals. On the other hand, cofinally many stages add a new real. In particular, the Cohen and Hechler stages add reals which are not in the preceding intermediate model, and hence yield ω_3 many distinct reals in the final model. Thus $\mathfrak{c} = \omega_3$. $\square$

Lemma 5.2. *The final extension satisfies $\mathfrak{b} = \omega_3$.*

Proof. Since $\mathfrak{b} \leq \mathfrak{c}$ and Lemma 5.1 gives $\mathfrak{c} = \omega_3$, it is enough to show that every family of functions in ω^ω of size $< \omega_3$ is bounded.

Let $F \subseteq \omega^\omega$ have size $\lambda < \omega_3$ in the final extension. Choose a name for an enumeration $\langle \dot{f}_i : i < \lambda \rangle$ of F . In a finite-support c.c.c. iteration, every name for a real has countable support: take a nice name, use that each antichain is countable, and collect the finitely many coordinates appearing in the conditions of those antichains. Hence the union of the supports of the names $\dot{f}_i$, $i < \lambda$, has size $< \omega_3$. By regularity of ω_3 it is bounded in ω_3 . Thus there is some $\beta < \omega_3$ such that $F \in L[G_\beta]$.

Choose $\alpha \in I_{\text{dom}}$ with $\alpha > \beta$. The Hechler real added at stage α dominates every function in $\omega^\omega \cap L[G_\alpha]$, hence dominates every member of F . This domination relation is absolute to all later forcing extensions. Therefore F is bounded in the final model. No family of size $< \omega_3$ is unbounded, and so $\mathfrak{b} \geq \omega_3$. The reverse inequality is $\mathfrak{b} \leq \mathfrak{c} = \omega_3$. $\square$

The maximal orthogonal family

Let $\mathcal{O}$ be the family consisting of the point measures together with the non-atomic measures inserted at successful I_{MO} stages. More precisely, when an I_{MO} stage sees a code f for a non-atomic measure μ_f which is orthogonal to the family constructed so far, it inserts the representative supplied by the Fischer–Friedman–Törnquist procedure and codes the corresponding MO-tagged successful-insertion record into the fresh MO-block assigned to that stage.

Proposition 5.3. *In the final extension, $\mathcal{O}$ is a Π_2^1 maximal orthogonal family of Borel probability measures on 2^ω .*

Proof. We first check orthogonality. The point measures are pairwise orthogonal, and a non-atomic measure is orthogonal to every point measure. Suppose a non-atomic measure is inserted at an I_{MO} stage α . The decision to insert it is made in $L[G_\alpha]$ only after the current measure has been checked to be orthogonal to all previous non-atomic members of the family. Orthogonality of coded Borel probability measures is absolute, by the Borelness recorded in Lemma 2.1. Moreover, the inserted representative is equivalent to the measure under consideration, so it has the same orthogonality relation to the earlier members. Thus the family is orthogonal at the moment of insertion, and later stages preserve the already established relations by the Fischer–Friedman–Törnquist preservation theorem recalled in Proposition 2.5.

For maximality, let f be a real in the final model coding a non-atomic Borel probability measure μ_f . Choose $\beta < \omega_3$ with $f \in L[G_\beta]$. By the local exhaustivity of the I_{MO} bookkeeping, some later stage $\alpha \in I_{\text{MO}}$ considers a name whose interpretation is f . If at that stage μ_f is orthogonal to the family already constructed, the construction inserts an equivalent measure into $\mathcal{O}$. Then μ_f is not orthogonal to that inserted member. If the stage does not insert such

a measure, the internal reason is that μ_f is already non-orthogonal to an earlier member of the family. In either case μ_f is non-orthogonal to some member of $\mathcal{O}$. The point measures cover the atomic case. Hence $\mathcal{O}$ is maximal.

It remains to justify that the projective definition has not changed. The original Fischer–Friedman–Törnquist definition says, in effect, that a measure code belongs to the constructed family iff every countable suitable model containing the code verifies a correct local successful-insertion pattern on an MO-block. This is a Π_2^1 condition by Proposition 2.5. Our extra stages use no MO-blocks and no MO-namespace successful-insertion payloads. The Σ -uniformization stages use only $\mathcal{B}^\Sigma$ and only U_{odd} - or U_{even} -payloads, the Cohen and Hechler stages are in the finite-support preservation class, and Lemma 3.9 prevents accidental Fischer–Friedman codes on unused coding coordinates and on unused namespace-tagged payloads. Therefore the same Π_2^1 formula defines exactly the intended family $\mathcal{O}$ in the mixed extension. $\square$

No low-complexity maximal orthogonal families

Proposition 5.4. *The final extension contains no boldface Σ_2^1 maximal orthogonal family of Borel probability measures.*

Proof. Let a be any real in the final extension. As in the proof of Lemma 5.2, the name for a has countable support and hence $a \in L[G_\beta]$ for some $\beta < \omega_3$. Choose $\alpha \in I_{\text{Cohen}}$ with $\alpha > \beta$. The real added at stage α is Cohen over $L[G_\alpha]$, and therefore Cohen over the inner model $L[a]$ because $L[a] \subseteq L[G_\alpha]$. Being Cohen over this fixed model is upward absolute to the rest of the iteration: every dense open subset of $2^{<\omega}$ coded in $L[a]$ is already present in $L[G_\alpha]$ and is met by the stage- α Cohen real. Proposition 2.3 now rules out $\Sigma_2^1(a)$ maximal orthogonal families. Since a was arbitrary, there are no boldface Σ_2^1 maximal orthogonal families. $\square$

Verification of global Σ -uniformization

Proposition 5.5. *The final extension satisfies UP_Σ .*

Proof. We verify the hypotheses of Proposition 4.5. The iteration is FF-respecting. Coordinates from $\mathcal{B}^\Sigma$ are used only at I_Σ stages, and every such stage either positively codes the U_{odd} - or U_{even} -payload prescribed by the current copying case or is trivial and does not intentionally code that payload. By Convention 3.2, these payloads are disjoint from all MO-tagged insertion records. Lemma 3.9 therefore gives positive coding, omission-based negative information, and no accidental codes for uniformization-namespace payloads.

The local candidate orders used in Section 4 are available at all initial segments. By Lemma 5.1, every proper initial segment has size $< \omega_3$, and the local continuum is below ω_3 . The locally cofinal partition and the dummy real-adding stages give the cardinal arithmetic needed for Lemma 4.1; hence the coherent maps π_ξ^γ exist for the initial segments of the present iteration. The support calculation from Lemma 4.2 applies because this is a finite-support c.c.c. iteration of regular length ω_3 .

Finally, the I_Σ bookkeeping is locally exhaustive for all data occurring in the four copying cases: projective formula indices, real parameters, inputs, local candidates, and the auxiliary compression reals representing sequences of earlier witnesses or later exclusions. Therefore every successful candidate is eventually processed at a stage at which the required compression real is already present in the intermediate model, and every earlier failure or later exclusion receives the prescribed coded tag or omitted tag. Proposition 4.5 applies and gives Σ_n^1 -uniformization for every $n \geq 3$. Kondo’s theorem gives the case $n = 2$. Thus the final extension satisfies UP_Σ . $\square$

Combining Lemmas 5.1 and 5.2 with Propositions 5.3, 5.4 and 5.5 proves Theorem 1.1.

6 The model with $\mathfrak{b} = \omega_1$ and $\mathfrak{c} = \omega_2$

We now prove Theorem 1.2. The construction is the countable-support analogue of the preceding section. The maximal-orthogonal-family part is again the Fischer–Friedman–Törnquist construction, but now in its S -proper, ω^ω -bounding form. The global Σ -uniformization module is inserted on a disjoint reservoir of Fischer–Friedman blocks, and random stages are used cofinally often to rule out boldface Σ_2^1 maximal orthogonal families.

The countable-support iteration

Work in L . Fix the stationary set $S \subseteq \omega_1$ and the definable Fischer–Friedman coding apparatus used in the countable-support constructions of [FFK14b, FFT12]. As before, the coding blocks are partitioned into two disjoint definable reservoirs

$$\mathcal{B}^{\text{MO}} \quad \text{and} \quad \mathcal{B}^\Sigma.$$

The first reservoir is used only by the maximal-orthogonal-family stages, and the successful-insertion definition reads only MO-namespace records. The second reservoir is used only by the uniformization stages, and the uniformization formulas from Section 4 read only U_{odd} - and U_{even} -records.

Let

$$J_{\text{MO}} \cup J_\Sigma \cup J_{\text{rand}} = \omega_2$$

be a partition into three cofinal sets. We choose it locally cofinal in the following sense: if $\gamma < \omega_2$ has cardinality ω_1 , then each of the three pieces has cardinality ω_1 below γ . This gives enough room in every uncountable initial segment for the maximal-family bookkeeping, for the uniformization bookkeeping, and for later random reals.

We build a countable-support iteration

$$\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha < \omega_2 \rangle.$$

The bookkeeping is the canonical L -bookkeeping, interpreted in the current intermediate model. It is chosen so that the following local exhaustivity condition holds.

For every $\beta < \omega_2$ and every object of the relevant kind in $L[G_\beta]$, there are cofinally many later stages of the appropriate type which reconsider that object, using a fresh block whenever a block is required.

Here the relevant objects are measure codes, projective formula indices, real parameters, local inputs, local candidates, and the auxiliary compression reals appearing in the four copying cases of Section 4. The iterands are assigned as follows.

- (1) If $\alpha \in J_{\text{MO}}$, the stage runs the countable-support Fischer–Friedman–Törnquist maximal-orthogonal-family step. It considers the next measure name selected by the bookkeeping. If, in $L[G_\alpha]$, the interpreted measure is non-atomic and orthogonal to the family built so far, the stage inserts the corresponding equivalent measure and records the successful insertion in a fresh block from $\mathcal{B}^{\text{MO}}$. If the measure is atomic or is already non-orthogonal to a previous member, the stage is trivial.
- (2) If $\alpha \in J_\Sigma$, the stage runs one step of the uniformization module. The task consists of a projective formula, a parameter, an input, a local candidate, and the finite list of auxiliary compression reals required by the relevant parity case. The stage either positively codes the prescribed U_{odd} - or U_{even} -payload using a fresh coordinate from $\mathcal{B}^\Sigma$,

or is trivial and does not intentionally code that payload. In the positive case the forcing used is the countable-support S -proper, ω^ω -bounding implementation of Lemma 3.9; in the negative case the forcing is trivial, and the later negative assertion is supplied by Lemma 3.8.

- (3) If $\alpha \in J_{\text{rand}}$, the iterand is random forcing, as computed in $L[G_\alpha]$.

The construction is FF-respecting in the sense of Definition 3.3. No stage using $\mathcal{B}^\Sigma$ is allowed to act on a block from $\mathcal{B}^{\text{MO}}$, and no stage using $\mathcal{B}^{\text{MO}}$ is allowed to act on a block from $\mathcal{B}^\Sigma$. In addition, the payload namespaces are kept disjoint: J_{MO} stages code only MO-records, while J_Σ stages code only U_{odd} - or U_{even} -records.

Lemma 6.1. *The countable-support iteration $\mathbb{P}_{\omega_2}$ preserves cardinals and stationary subsets of ω_1 , has size ω_2 , and forces $\mathfrak{c} = \omega_2$. Moreover, every proper initial segment $\mathbb{P}_\gamma$, $\gamma < \omega_2$, has size at most ω_1 and preserves CH.*

Proof. The non-trivial iterands at J_{MO} stages are the S -proper iterands from the Fischer–Friedman–Törnquist countable-support construction. The non-trivial iterands at J_Σ stages are S -proper by Lemma 3.9. Random forcing is proper, and in the present context it is S -proper as well: the generic Borel positive set can be chosen to meet the usual dense sets belonging to any countable model $M \prec H_\theta$ with $M \cap \omega_1 \in S$. Therefore the whole countable-support iteration is S -proper by the standard preservation theorem.

The same preservation theorem is the one used in [FFK14b, FFT12]: under CH, a countable-support iteration of length at most ω_2 whose iterands have size at most ω_1 and are S -proper in the required sense preserves ω_1 , preserves the stationarity of S , and preserves ω_2 . Equivalently for the present argument, there is no collapse of cardinals in the Fischer–Friedman countable-support implementation, and adding the disjoint Σ -reservoir stages does not change the hypotheses of the preservation theorem.

We record the size calculation because it is used again in the compression argument. For $\gamma < \omega_2$ we have $|\gamma| \leq \omega_1$. Since L satisfies CH and hence $(\omega_1)^\omega = \omega_1$, the set of countable supports contained in γ has size at most ω_1 . By induction on γ , the iterands appearing below γ have size at most ω_1 , and therefore $|\mathbb{P}_\gamma| \leq \omega_1$. Thus a proper initial segment adds at most ω_1 many reals. Since it also preserves ω_1 , CH holds in $L[G_\gamma]$.

For the full iteration, the same calculation with ω_2 in place of γ gives $|\mathbb{P}_{\omega_2}| = \omega_2$, using GCH in L . Hence the final extension has at most ω_2 many reals. Conversely, cofinally many J_{rand} stages add reals which are not in the previous intermediate model. Thus the final continuum is exactly ω_2 . $\square$

Lemma 6.2. *The final extension satisfies $\mathfrak{b} = \omega_1$.*

Proof. The iterands at J_{MO} and J_Σ stages are ω^ω -bounding by the Fischer–Friedman preservation theorem and by Lemma 3.9. Random forcing is ω^ω -bounding. Hence the whole countable-support iteration is ω^ω -bounding by the standard preservation theorem for countable-support iterations of proper ω^ω -bounding forcings.

Let $B \subseteq \omega^\omega \cap L$ be an L -unbounded family of size ω_1 . We claim that B remains unbounded. Suppose, towards a contradiction, that some final function $g \in \omega^\omega$ dominates every member of B . Since the forcing is ω^ω -bounding, there is an old function $h \in \omega^\omega \cap L$ with $g \leq^* h$. Then h dominates every member of B , contradicting the choice of B in L . Thus $\mathfrak{b} \leq \omega_1$ in the final extension. The reverse inequality $\mathfrak{b} \geq \omega_1$ is absolute for forcing extensions preserving ω_1 . Therefore $\mathfrak{b} = \omega_1$. $\square$

The maximal orthogonal family

Let $\mathcal{O}$ be the family consisting of the point measures together with the non-atomic measures inserted at successful J_{MO} stages. Membership in the non-atomic part is read only from successful Fischer–Friedman–Törnquist insertion records on blocks from $\mathcal{B}^{\text{MO}}$, with the corresponding payloads in the MO-namespace.

Proposition 6.3. *In the final extension, $\mathcal{O}$ is a Π_2^1 maximal orthogonal family of Borel probability measures on 2^ω .*

Proof. The proof is the same Fischer–Friedman–Törnquist verification as in Proposition 2.5, with the reservoir separation made explicit. When a measure is inserted at a J_{MO} stage, the current model has verified that it is orthogonal to all earlier non-atomic members. Orthogonality of coded Borel probability measures is absolute by Lemma 2.1, and the inserted representative is equivalent to the measure under consideration. Thus the family is orthogonal when the insertion is made, and later S -proper preservation keeps the local coding record correct.

For maximality, let f be a real in the final extension coding a Borel probability measure μ_f . If μ_f has an atom, it is non-orthogonal to a point measure. Assume therefore that μ_f is non-atomic. By the support-localization lemma for the present countable-support iteration, Lemma 4.2, there is $\beta < \omega_2$ with $f \in L[G_\beta]$. By local exhaustivity, a later stage in J_{MO} considers a name for this same measure. If the measure is still orthogonal to the family built so far, the stage inserts an equivalent measure into $\mathcal{O}$; otherwise it is already non-orthogonal to an earlier member of $\mathcal{O}$. In both cases μ_f is non-orthogonal to some member of $\mathcal{O}$. Hence $\mathcal{O}$ is maximal.

The projective definition is unchanged by the added uniformization and random stages. The original definition of the non-atomic part of $\mathcal{O}$ is the Fischer–Friedman–Törnquist Π_2^1 successful-insertion predicate, read on MO-blocks and MO-namespace insertion records. Stages in J_Σ use only $\mathcal{B}^\Sigma$ and only U_{odd} - or U_{even} -payloads, random stages do not use Fischer–Friedman blocks, and Lemma 3.9 rules out accidental codes on unused MO-coding coordinates or unused MO-namespace insertion records. Therefore the same Π_2^1 formula defines exactly $\mathcal{O}$ in the mixed extension. $\square$

No low-complexity maximal orthogonal families

Proposition 6.4. *The final extension contains no boldface Σ_2^1 maximal orthogonal family of Borel probability measures.*

Proof. Let a be an arbitrary real in the final extension. By the one-real case of Lemma 4.2, there is $\beta < \omega_2$ with $a \in L[G_\beta]$. Choose $\alpha \in J_{\text{rand}}$ with $\alpha > \beta$. The real added by the random forcing at stage α is random over $L[G_\alpha]$, and hence random over the inner model $L[a]$.

This remains true in the final extension in the only sense needed here. A Borel null set coded in $L[a]$ is already coded in $L[G_\alpha]$, and the stage- α random real avoids it. Later forcing cannot turn this fixed old Borel set into a set containing the random real. Therefore the final extension contains a random real over $L[a]$. Proposition 2.3 rules out $\Sigma_2^1(a)$ maximal orthogonal families. Since a was arbitrary, there are no boldface Σ_2^1 maximal orthogonal families. $\square$

Verification of global Σ -uniformization

Proposition 6.5. *The final extension satisfies UP_Σ .*

Proof. We check the hypotheses of Proposition 4.5. The iteration is FF-respecting. Coordinates from $\mathcal{B}^\Sigma$ are used only at J_Σ stages, and each such stage implements exactly one of the

four copying cases from Section 4: either it codes the prescribed U_{odd} - or U_{even} -payload using a fresh Σ -coding coordinate, or it is trivial and does not intentionally code the corresponding payload. By Convention 3.2, these payloads are disjoint from all MO-tagged insertion records. By Lemma 3.9, positive coding, omission-based negative information, and no accidental coding hold for the S -proper implementation.

The local compression maps from Lemma 4.1 are available because proper initial segments have size at most ω_1 and preserve CH by Lemma 6.1. Thus, in every $L[G_\gamma]$ with $\gamma < \omega_2$, all local predecessor sequences of length below the local continuum can be compressed into single reals, and the maps can be chosen coherently as the initial segment grows. The localization lemma, Lemma 4.2, applies to the present countable-support iteration because a sequence of length $< \omega_2$ uses fewer than ω_2 many coordinates and hence is already interpreted in a bounded initial segment.

Finally, the J_Σ bookkeeping is locally exhaustive for the data used by the uniformization module. Once the least successful candidate for a projective section has appeared in an intermediate model, the finitely many auxiliary compression reals needed to certify the earlier failures or later exclusions can be localized to a later intermediate model. A still later J_Σ stage then presents exactly that task and uses a fresh Σ -coding coordinate to impose the prescribed coded uniformization payload, or leaves that payload out of the intentionally coded set. Therefore Proposition 4.5 gives Σ_n^1 -uniformization for every $n \geq 3$. Kondo's theorem gives the case $n = 2$. Hence the final model satisfies UP_Σ . $\square$

Combining Lemmas 6.1 and 6.2 with Propositions 6.3, 6.4 and 6.5 proves Theorem 1.2.

7 Concluding remarks

The proofs above deliberately separate the maximal-orthogonal-family construction from the uniformization construction. This separation is useful for two reasons. First, it makes clear that the original Fischer–Friedman–Törnquist Π_2^1 definition of the maximal orthogonal family is not disturbed by the new uniformization stages. Second, it shows that the global Σ -uniformization argument does not depend on the special coding predicates used in earlier global-uniformization papers; any coding package with the formal features of Lemma 3.9 can be used as the base predicate.

The same reservoir-and-namespace argument also allows one to reserve a further disjoint class of coding coordinates and a further disjoint payload namespace, and run the original Fischer–Friedman wellorder bookkeeping in parallel. This yields the corresponding Δ_3^1 wellorder of the reals together with the conclusions of Theorems 1.1 and 1.2, in line with [FFK14b, FFT12, Hof25c, Hof25d]. Since the wellorder is not needed for the uniformization argument presented here, we have kept it out of the statement of the main theorems.

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